Tuning report: tropical / FractionalKnapsack

2026-10-22T19:17:01

1 Decision

Approving freezes `fishr.lambda` = 10 into every cell of the grid (frozen hash `05d2df6e56c7`). The experiment YAML has `fishr.lambda` = 1000 for this campaign. `frozen.yaml` is currently unapproved with {'fishr.lambda': 10.0}.

Selection rule. The candidate with the lowest mean `id_val_loss` wins, averaged over 6 cell(s) and 2 tuning seed(s) each (global seed 1000, disjoint from the grid's 0), across 4 candidates. A candidate with a missing or diverged run is not eligible.

Verdict: REVIEW

- **review**: fishr.lambda=10 is the smallest value searched: the optimum may lie beyond it, so widen the space
- **review**: 12 of 12 winner runs reached their best id_val_loss at the last logged epoch: the search budget may be too short to rank candidates
- **review**: 12 of 12 winner runs picked their last epoch (best_epoch = 19): still improving when the search budget ran out, so candidates were ranked before converging
- **info**: the search ran with optim.epochs=20; the grid runs with optim.epochs=100
- **info**: devices: cuda:0

block: do not approve; fix the search and re-run it. **review**: read the sections below before approving. **info**: context only.

2 Search setup

- `fishr.lambda` ∈ {10, 100, 1000, 1e+04}
- Cells trained (`tune.only` {'objective': 'fishr'}):
  - `model_type=vanilla head=seq2seq objective=fishr`
  - `model_type=vanilla head=pointer objective=fishr`
  - `model_type=adaptive head=seq2seq objective=fishr`
  - `model_type=adaptive head=pointer objective=fishr`
  - `model_type=tropical head=seq2seq objective=fishr`
  - `model_type=tropical head=pointer objective=fishr`
- Search-only overrides (never frozen): `{'optim': {'epochs': 20}}`. The chosen value was ranked under this budget, not the grid's.

3 Candidates

Table 1 ranks the candidates on `id_val_loss` (lower is better); Figure 1 plots them. Candidates share seeds, so Table 2 compares each with the winner run by run (same cell, same seed).

Table 1: Every candidate: rank, runs stored/expected, diverged, mean with 95% bootstrap CI, SD and range over all its runs.

#	`fishr.lambda`	runs	div.	mean	95% CI	SD	range	
1	10	12/12	0	0.2813	[0.249, 0.312]	0.0591	[0.189, 0.354]	**selected**
2	100	12/12	0	0.2848	[0.256, 0.313]	0.0555	[0.197, 0.354]	
3	1000	12/12	0	0.2974	[0.274, 0.32]	0.0423	[0.225, 0.354]	
4	1e+04	12/12	0	0.3036	[0.281, 0.324]	0.0391	[0.24, 0.354]	

Table 2: Each candidate minus the winner on the runs they share, signed so a positive gap is worse than the winner. Worse/better/equal count the runs; p is a two-sided sign test.

candidate	n	gap	95% CI	worse	better	equal	p
`fishr.lambda=100`	12	0.00346	[0.00142, 0.00589]	7	5	0	0.7744
`fishr.lambda=1000`	12	0.01608	[0.00685, 0.0258]	9	3	0	0.1460
`fishr.lambda=10000`	12	0.02233	[0.0114, 0.0346]	11	1	0	0.0063

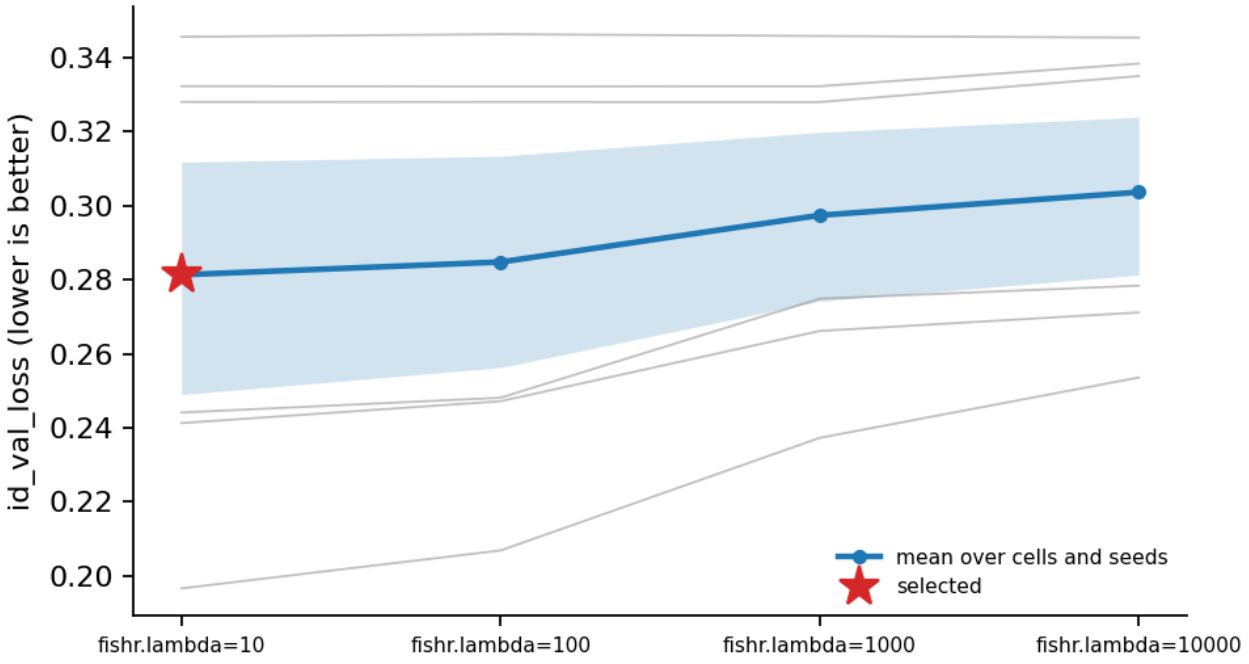


Figure 1: `id_val_loss` per candidate: the mean over every run with a 95% bootstrap CI (blue), each cell's seed-mean (grey), the selected candidate (red star).

4 Per cell

One value is frozen for every cell, chosen by the mean over cells. Figure 2 shows whether the winner is best in each cell or is carried by a few of them.

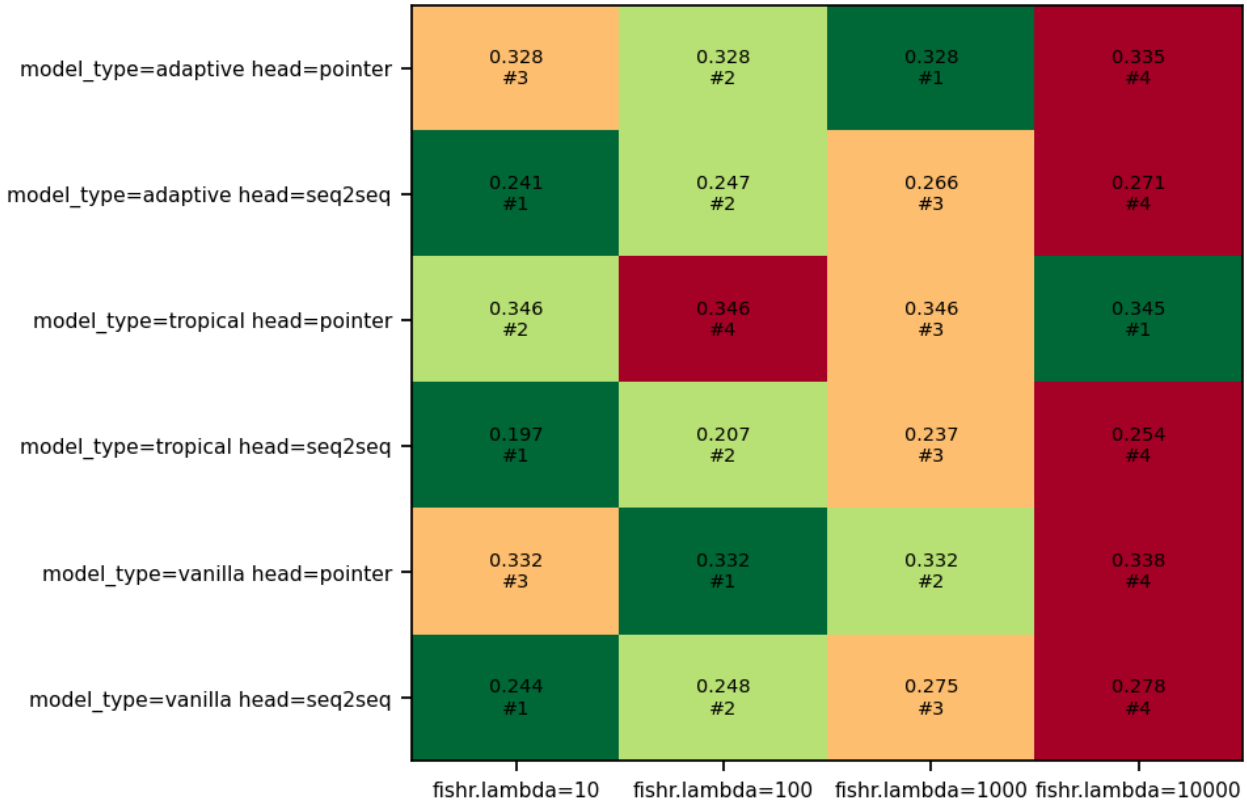


Figure 2: Each cell's seed-mean `id_val_loss` per candidate, with its rank within the cell (#1 is best; green is better).

Table 3: The best candidate in each cell.

cell	best
`model_type=adaptive head=pointer`	fishr.lambda=1000
`model_type=adaptive head=seq2seq`	fishr.lambda=10
`model_type=tropical head=pointer`	fishr.lambda=10000
`model_type=tropical head=seq2seq`	fishr.lambda=10
`model_type=vanilla head=pointer`	fishr.lambda=100
`model_type=vanilla head=seq2seq`	fishr.lambda=10

5 Other training-distribution metrics

Not used for selection; shown so a winner that trades them away is visible.

Table 4: Mean of every other in-distribution metric per candidate.

candidate	`id_f1`	`id_exact`	`fishr_penalty_final`	`best_epoch`
fishr.lambda=10	0.922	0.483	0.00101	19

Table 4: Mean of every other in-distribution metric per candidate.

candidate	id_f1	id_exact	fishr_penalty_final	best_epoch
fishr.lambda=100	0.921	0.477	3.23×10^{-4}	19
fishr.lambda=1000	0.916	0.456	7.74×10^{-5}	19
fishr.lambda=10000	0.915	0.447	3.70×10^{-5}	12.5

6 Training curves

The mean over every cell and seed of each candidate; the selected candidate in bold red.

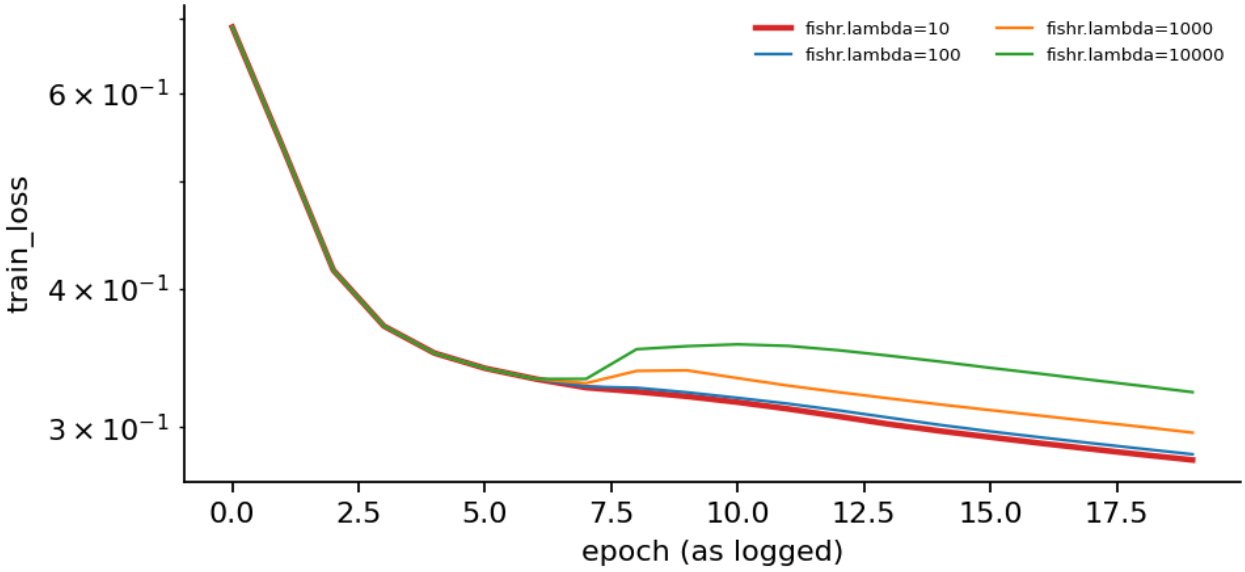


Figure 3: `train_loss` per candidate.

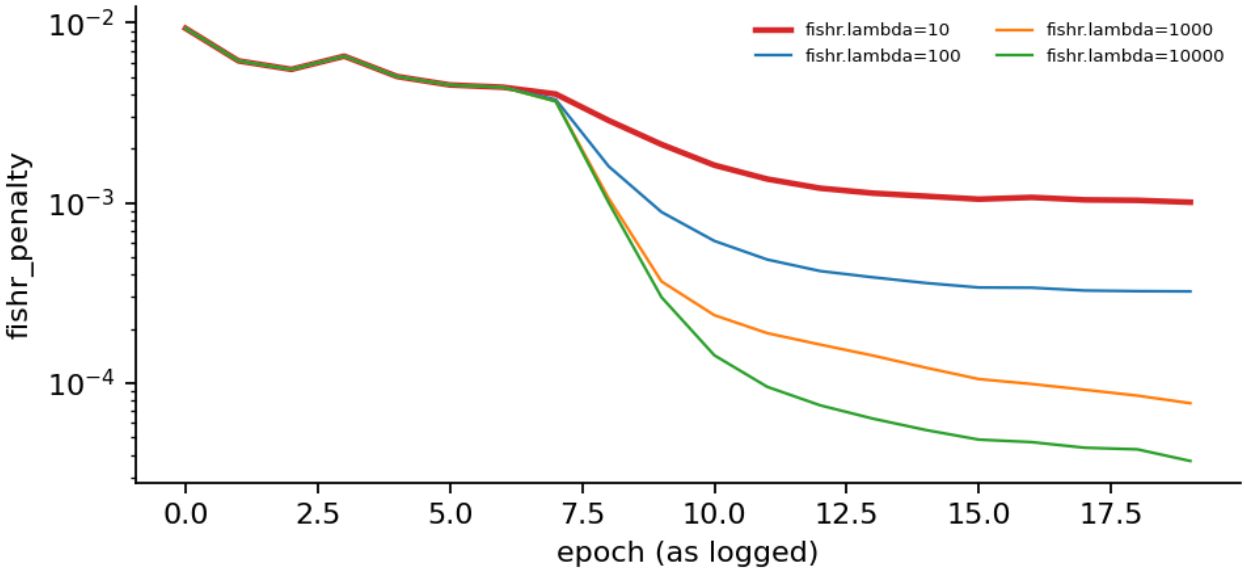


Figure 4: `fishr_penalty` per candidate.

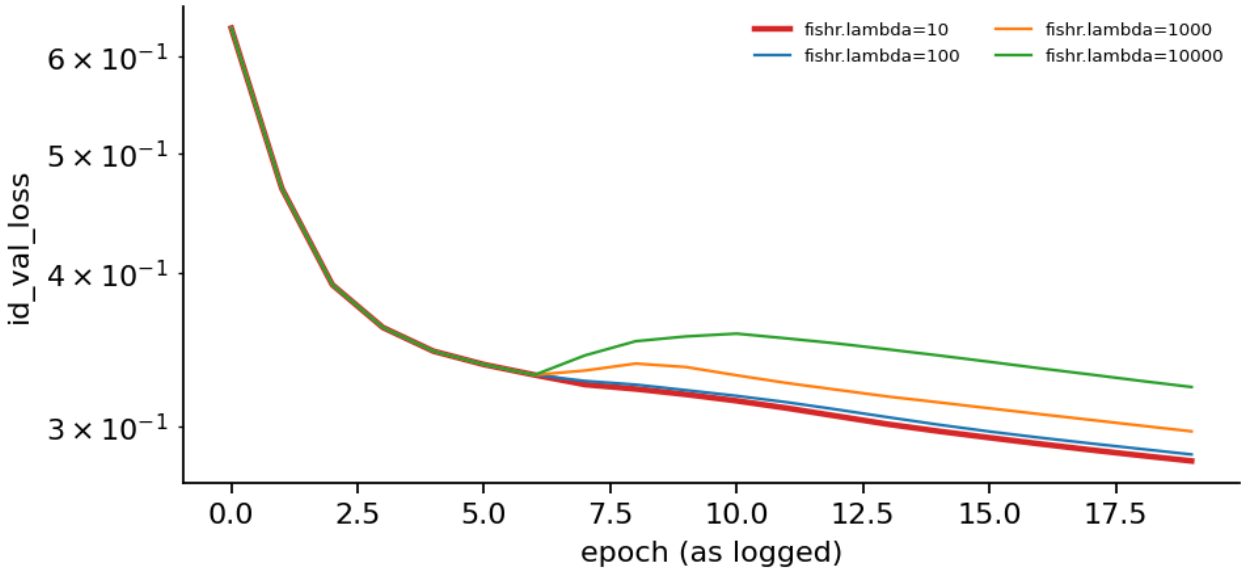


Figure 5: `id_val_loss` per candidate.

7 Provenance

Table 5: Each candidate's runs: base hash, runs, total wall time, commits (+ marks a dirty tree), torch.

candidate	hash	runs	wall (s)	commit	torch
fishr.lambda=10	`1edaf7a77ea6`	12	3.61e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=100	`a00feda41def`	12	3.62e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=1000	`37ffd00e8792`	12	3.62e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=10000	`6bbd32f95f94`	12	3.62e+03	`e5ccce8034`	2.8.0+cu128

Store: `campaigns/tropical/FractionalKnapsack/tune/results`.
Rebuild: `expkit tune-report --config configs/tropical.yaml --profile FractionalKnapsack`
Approve: `expkit approve --config configs/tropical.yaml --profile FractionalKnapsack`

The base the grid would run with.

```
task:
  name: FractionalKnapsackDataset
  value_range:
  - 1
  - 10
  target_range:
  - 10
  - 20
  weight_range:
  - 1
  - 10
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 1
  - 5
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
```

```
lambda: 10.0
ema: 0.95
anneal_iters: 1500
reset_optimizer: true
```

A Sealed: OOD and test metrics

Read this only after deciding. These were recorded by the tuning runs but never consulted by the selection. Choosing a value because it looks better here would tune on the test distribution.

Table 6: Mean of every OOD or test metric per candidate.

candidate	ood_len_f1	ood_len_exact	ood_val_f1	ood_val_exact	ood_noise_f1	ood_noise_exact	ood_len_f1_L12	ood_len_exact_L12	ood_len_f1_L16	ood_len_exact_L16	ood_len_f1_L64	ood_len_exact_L64
fishr.lambda=10	0.585	0.0439	0.871	0.3	0.843	0.235	0.839	0.105	0.77	0.0263	0.45	0
fishr.lambda=100	0.58	0.0406	0.871	0.3	0.843	0.233	0.835	0.0996	0.766	0.0221	0.444	0
fishr.lambda=1000	0.569	0.0365	0.864	0.281	0.846	0.237	0.825	0.0916	0.755	0.0179	0.434	0
fishr.lambda=10000	0.573	0.0387	0.865	0.279	0.85	0.247	0.831	0.0983	0.761	0.018	0.437	0